

Kalhara Tennakoon

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PROFESSIONAL SUMMARY

First Class Computer Science (Software Engineering) graduate and batch top with hands-on industry experience in software development, cloud-native systems, Kubernetes, CI/CD automation, and cloud security at IFS R&D International. Experienced in building full-stack applications, REST APIs, ML forecasting systems, and AI applications using local LLMs, RAG, vector search, and PostgreSQL. Strong foundation in Python, JavaScript, React, Node.js, Java, Go, databases, cloud platforms, and modern DevOps practices. Interested in building reliable, scalable, and intelligent software systems across Software Engineering, Cloud, DevOps, and MLOps.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, Go, C, Bash/Shell Scripting

Software Engineering: React, Node.js, Express, Flask, FastAPI, REST APIs, Full-Stack Development, API Development, JWT Authentication

AI / Machine Learning: Scikit-learn, Pandas, Prophet, pgvector, Ollama, RAG, Local LLMs, Data Cleaning, Exploratory Data Analysis, Feature Engineering, Time-Series Forecasting

Cloud & DevOps: Kubernetes, Docker, Tekton, Argo CD, Kustomize, GitHub Actions, Azure DevOps, Trivy, CI/CD, Infrastructure as Code, GitOps, Container Security

Cloud Platforms: Microsoft Azure, AWS, Google Cloud Platform, AKS, GKE, Rancher

Databases: PostgreSQL, MySQL, MongoDB

Tools & Practices: Git, GitHub, Jira, Agile, Scrum, Technical Documentation, Automated Testing

PROFESSIONAL EXPERIENCE

IFS R&D International (Pvt) Ltd

Colombo, Sri Lanka

DevOps Engineering Intern

Jul 2024 – Oct 2024

- Automated Kubernetes-native CI/CD workflows using Tekton, streamlining deployment processes for critical services.
- Integrated Behave automated testing into Tekton CI/CD pipelines, reducing manual debugging effort by approximately 30% and improving deployment reliability.
- Optimized pipeline configurations and resolved recurring build and deployment issues to improve deployment stability and engineering efficiency.
- Conducted knowledge-sharing sessions on testing best practices to support team collaboration and code quality.

Software Engineering Intern

Jul 2020 – Dec 2020

- Developed a Docker and Kubernetes-based security scanning solution for container platforms and vulnerability assessment workflows.
- Integrated security scanning tools into CI/CD pipelines to improve vulnerability detection across deployment stages.
- Diagnosed deployment and runtime issues and authored technical documentation supporting system reliability and onboarding.
- Worked with cloud-native technologies and containerized environments to strengthen secure software delivery practices.

PROJECTS

VetCare One — Smart Veterinary Clinic Management System

React, Node.js, Express, PostgreSQL, Python, Flask, Scikit-learn, pgvector, Ollama, SwiftUI

Oct 2025 – Aug 2026

- Built a full-stack veterinary clinic management platform covering appointments, medical records, billing, inventory, role-based access control, and JWT authentication.
- Developed a locally hosted RAG assistant using Ollama and pgvector, combining semantic retrieval with exact SQL queries for reliable access to structured clinic data.
- Implemented human-in-the-loop controls requiring confirmation for database write operations.
- Developed ML models for disease prediction, sales forecasting, and inventory demand forecasting.
- Built a pet-owner portal and SwiftUI companion iOS application to extend access beyond the clinic platform.
- Selected among the Top 10 finalists of the Ascentic AI Launch Pad 2026.

SolarCast — Solar Energy Forecasting Application with GitOps Deployment

Python, FastAPI, React, Prophet, Docker, Kubernetes, Kustomize, Argo CD, GitHub Actions

Apr 2026 – Oct 2026

- Built a full-stack machine learning application with a FastAPI backend and React dashboard that forecasts monthly solar generation, grid export, and expected payouts from residential inverter data.
- Developed a time-series forecasting pipeline using Meta Prophet with 90% confidence ranges, tested on unseen months with a chronological 70/30 split: about 8% error (MAPE) against 9.5% for a same-month-last-year baseline.
- Built a GitHub Actions CI/CD pipeline with linting, 39 automated tests, Trivy vulnerability scans, and end-to-end smoke tests, publishing multi-stage, non-root Docker images to Docker Hub.

- Deployed to Kubernetes dev and prod environments using Kustomize and Argo CD: merges to main roll out to dev, prod is promoted by pull request, and rollbacks are Git reverts.

Predicting Course Difficulty from Student Evaluation Responses

Python, Scikit-learn, Pandas, Matplotlib, Seaborn, SPSS

Feb 2025 – Jul 2025

- Developed supervised machine learning models to predict perceived course difficulty using student evaluation datasets.
- Performed data preprocessing, exploratory data analysis, feature engineering, and model evaluation using cross-validation.
- Applied statistical significance testing and model interpretation to identify instructor-related attributes associated with perceived course difficulty.

Kubernetes Cluster & Container Image Security Scanner

Kubernetes, Docker, Go, Argo CD, Rancher, Azure DevOps

Jul 2020 – Dec 2020

- Developed a containerized security scanning solution for Kubernetes environments to detect vulnerabilities in container images and cluster configurations.
- Integrated automated security scanning workflows into CI/CD pipelines to strengthen secure deployment practices and vulnerability detection coverage.
- Utilized cloud-native platforms and GitOps tooling to support scalable, secure, and reliable Kubernetes operations.

EDUCATION

Kingston University, London, United Kingdom

BSc (Hons) Computer Science (Software Engineering) — First Class Honours

2026

- Batch Top – Software Engineering, ESU Kandy

CERTIFICATIONS

Cloud, DevOps & Platform Engineering

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Azure Data Fundamentals (DP-900)
- Introduction to Kubernetes — The Linux Foundation
- Containers & Kubernetes Essentials — IBM
- GitHub Foundations — GitHub
- Introduction to GitOps

Artificial Intelligence & Data

- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Google AI Professional Certificate

HONOURS & AWARDS

Top 10 Finalist – Ascentic AI Launch Pad 2026

Selected among 700+ applicants and advanced to the final round.

Batch Top – Software Engineering, ESU Kandy 2026

Ranked first in the Software Engineering cohort.

Two-Time Dan Kohn Scholarship Recipient – KubeCon + CloudNativeCon Europe 2022 & 2025

Awarded scholarships recognizing contributions to the cloud-native ecosystem.

First Runner-Up – Hack:Bit Hackathon 2019

Secured 2nd place in a competition organized in collaboration with Microsoft and Sarvodaya Fusion.

Top 15 Finalist – HaXmas Hackathon 2018

Selected among 75+ competing teams.

Cisco NetRiders / Packet Riders

Ranked 4th in Sri Lanka and 118th in the Asia Pacific & Japan region.

TECHNICAL WRITING

Technical Writer — Medium

Dec 2021 – Present

- Authored 50+ technical articles covering DevOps, cloud-native technologies, Kubernetes, Tekton, Docker, Go, CI/CD, and software engineering.
- Generated more than 150,000 views across published technical content.
- Published technical content through platforms including Level Up Coding and FAUN.dev.

COMMUNITY ENGAGEMENT

Volunteer — STEMUp Educational Foundation

Aug 2018 – Dec 2020

- Contributed to initiatives promoting STEM education and digital literacy among students and youth communities.